

REVIEW

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A Narrative Review of Effects and Underlying Mechanisms of Needling Therapy for Treating Myofascial Pain and Musculoskeletal Disorders

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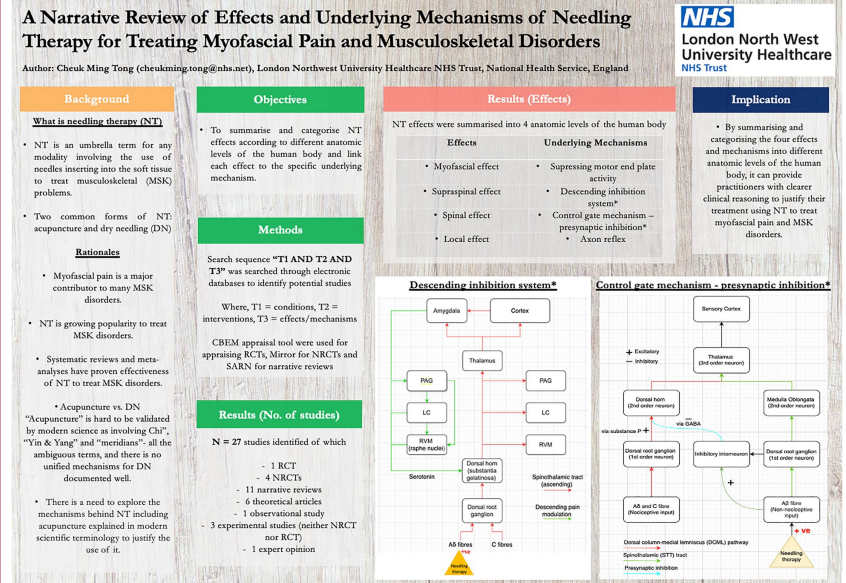


Abstract

This review aims to summarise the effects and underlying mechanisms of needling therapy (NT) in the treatment of myofascial pain and musculoskeletal (MSK) disorders. This review aims doing so by categorising NT effects according to different anatomic levels of the human body and linking each effect to the specific mechanism. Acupuncture, the first form of NT, has existed for 3000 years and been practised by Traditional Chinese Medicine (TCA); however, its mechanism is hard to be validated by modern science as it involves “Chi”, “Yin & Yang” and “meridians”—all the ambiguous concepts which are hard to be interpreted by the West and modern science. Dry needling (DN), another form of NT, which emerged in the West in the twentieth century, has not well been explicitly reviewed for its mechanisms. Given that NT—acupuncture originated in the East and DN originated in the West—has been proven by systematic reviews and meta-analyses for its effectiveness to treat MSK disorders (world’s major disability contributor as defined by World Health Organization) and that many MSK disorders are caused by myofascial pain, a summary of NT effects and mechanisms in modern scientific terminology can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders. This review summarised 4 main effects according to different anatomic levels of the human body attributed to NT. They are myofascial effect, supraspinal effect, spinal effect and local effect. The mechanisms responsible for each of these effects, respectively, are suppressing motor end plate activity, descending inhibition mechanism, control gate mechanism and axon reflex.

Keywords Needling therapy, Acupuncture, Dry needling, Myofascial pain, Musculoskeletal disorders, Mechanisms, Effects

Graphical Abstract



1 Introduction

Musculoskeletal (MSK) disorders are defined as conditions affecting muscles, ligaments, tendons, bones and nerves acutely and chronically [1]. Data from the Global Burden of Disease shows that MSK disorders are the most prevalent category of health issue affecting nearly 1.71 billion people worldwide [2]. In the United Kingdom, almost one third of the population suffer from MSK disorders [3]. The World Health Organization has defined MSK disorders as the leading contributor to disability worldwide [4]. While the most prevalent MSK disorders is low back pain [4], 85% of low back pain is caused by myofascial pain [5], and myofascial pain accounts for 30–85% of other different MSK disorders [6]. A systematic review conducted by Babatunde et al. [7] has shown that exercise therapy, psychosocial interventions and pharmacological treatments are effective to treat MSK disorders. However, these commonly

practised modalities appear to be insufficient to lower the high prevalence of MSK disorders.

Needling therapy (NT), therefore, starts to gain its popularity as a new modality to treat MSK disorders due to growing evidence to prove its efficacy. NT is an umbrella term for any modality involving the use of needles inserting into the soft tissue to treat MSK pain and conditions [8]. There are two common forms of NT: acupuncture and dry needling (DT). While DT is defined by the American Physical Therapy Association (APTA) as a skilled intervention involving the use of a filiform needle to penetrate the skin to stimulate underlying myofascial points, muscular and connective tissues to treat MSK pain and movement impairments [9], acupuncture (translated as needle penetration) is defined in TCM as the use of small needles inserted into the specific points of the body (acupoints points) to treat bodily problems [10]. Two systematic reviews have shown effectiveness

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